



InnoScream bot

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PROJECT OVERVIEW

InnoScream: Anonymous Stress Relief Platform for Students

- Telegram-based platform for students to vent frustrations anonymously
- Reactions, memes, and weekly analytics promote community support
- Developed in 1 week by a team of 5
- FastAPI backend + SQLite + aiogram bot



CORE FUNCTIONALITY



Key Features

1. **/scream [text]** — anonymous post submission
2. **Reactions:** 🥹 🔥 🤡
3. **Top Screams:** Daily most-voted posts get pinned and memefied
4. **Weekly Stress Graphs:** Visualization via QuickChart.io
5. **/stats:** Personal post count (anonymous, hashed tracking)
6. **Moderation:** Admins delete with /delete [id] or reply with /delete

PROJECT SCOPE AND ADJUSTMENTS

- Memes generated once per day for top post
- Switched from ImgFlip to Unsplash API for more flexible image generation
- Admin moderation improved: reply with /delete for ease of use
- Statistics and memes published automatically to the channel
- Emoji availability constraints handled (💀 → 😭)





QUALITY REQUIREMENTS



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MAINTAINABILITY

Flake8: 0 warnings

Docstrings for all functions
(pydocstyle)

Code complexity ≤ 10 (Radon)

RELIABILITY

Test coverage $\geq 60\%$
(pytest-cov)

Mutation testing survival rate
 $\geq 80\%$ (mutmut)



PERFORMANCE

Bot response time $\leq 500\text{ms}$
under 100 RPS (Locust)

SQL query time $\leq 50\text{ms}$

SECURITY

No high-severity
issues (Bandit)

Data anonymity via
user_id hashing

AUTOMATION AND CI/CD

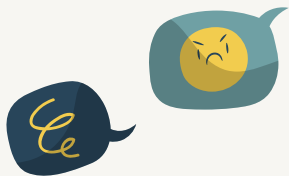


- Pre-commit checks (Flake8, pydocstyle)
- GitHub Actions for CI
- Auto-tests + linting + coverage
- Ensures continuous quality during fast-paced dev

LESSONS LEARNED

- Negotiating feature scope early avoids tech debt later
- Choosing APIs with flexible output saves dev time
- Pre-commit + CI tooling helps maintain standards under tight deadlines
- Team coordination and clear task distribution are crucial for 1-week sprint





TECH STACK RECAP

- **Backend:** FastAPI, Python 3.11
- **Bot Framework:** aiogram
- **Database:** SQLite
- **APIs:** Unsplash (memes), QuickChart.io (graphs)
- **Quality Tools:** pytest, Flake8, Bandit, Locust, mutmut, Radon



THANK YOU FOR YOUR ATTENTION



Project repository link:
<https://github.com/BugaevGleb/InnoScream>